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ORIGINAL ARTICLE

A holistic strategy for choosing software requirements through multi-criteria intuitionistic fuzzy TOPSIS

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Abstract
Selecting the most suitable software requirements is crucial for project success, but remains challenging due to inherent uncertainty, complexity, and conflicting evaluation criteria. Traditional prioritization approaches often fail to capture hesitation in decision-making, leading to suboptimal outcomes adequately. To address this, we propose a novel decision-making framework that integrates Intuitionistic Fuzzy Sets (IFS) with the Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS). The approach effectively models membership, non-membership, and hesitation degrees, thereby providing a richer representation of uncertainty. A structured methodology is presented, including criteria definition, weight determination, intuitionistic fuzzy decision matrix construction, and TOPSIS-based ranking. A case study on an Institute Examination System demonstrates the framework's applicability and effectiveness, showing consistent ranking results with reduced execution time compared to Fuzzy TOPSIS. The proposed method offers a more accurate and efficient mechanism for software requirements selection, supporting informed and reliable decision-making.

Keywords Software requirements selection · Intuitionistic fuzzy sets · TOPSIS · Multi-criteria intuitionistic fuzzy TOPSIS

1 Introduction

The software development process includes a pivotal stage known as software requirements selection, wherein the most appropriate collection of requirements is chosen for inclusion in a software project. It comprises identifying, prioritizing, and evaluating several software requirements based on their significance, feasibility, and alignment with the objectives of the project (Pitangueira et al. 2015). The aim of the process of requirements selection is to select the most appropriate requirements because it is impossible to consider all requirements in a single version of any software product due to various constraints of a company; and to ensure that the selected requirements perfectly capture the needs and expectations of stakeholders while considering resource limits and project constraints (Nazim et al. 2022a). Appropriate requirements selection necessitates a systematic approach, involving collaboration between stakeholders, detailed analysis of requirements, and consideration of several aspects such as scalability, performance, regulatory compliance, functionality, usability, etc. (Nazim et al. 2021, 2022b). By selecting the appropriate set of requirements, software projects can improve their chances of success, reduce risks, and produce software solutions that satisfy the needs of users and the intended goals (Sadiq et al. 2022). A software requirement is selected and assessed using diverse criteria. Consequently, the act of choosing software requirements is commonly approached as a "*Multi-Criteria Decision-Making* (MCDM)" challenge (Nazim et al. 2022a).

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MCDM methods are useful tools used to support decision-making processes when dealing with complicated problems involving several criteria or objectives. These methods offer a structured way to evaluate and rank alternatives according to how well they perform against various criteria. The goal of MCDM methods is to facilitate the selection of the best alternative by considering a variety of perspectives, preferences, and trade-offs between different criteria (Memari et al. 2019). MCDM methodologies encompass a broad spectrum of strategies such as TOPSIS, "*Elimination and Choice Expressing Reality* (ELECTRE)," "*Analytic Hierarchy Process* (AHP)," and "*Preference Ranking Organization Method for Enrichment Evaluations* (PROMETHEE)" (Achimugu et al. 2014a). By employing MCDM techniques, decision-makers can more efficiently assess choices, enhance clarity, and improve decision-making across various domains, including business, public policy, and engineering. In the field of software engineering, MCDM approaches have been extensively utilized to streamline the process of selecting software requirements.

According to the latest literature review on SR selection by Nazim et al. (Nazim et al. 2021), the MCDM methods are one of the prevalent techniques for the selection of SRs. In most cases, the MCDM methods are also used under a fuzzy environment. The concept of fuzzy is applied because of the imprecision of human judgments for the selection of SRs based on various selection criteria. Previous literature exclusively employed conventional fuzzy numbers; nonetheless, utilizing *Intuitionistic Fuzzy Numbers* (IFNs) can yield superior outcomes due to their capacity to offer a more comprehensive and accurate depiction of uncertainty and imprecision in decision-making. IFNs consider both membership degrees and non-membership degrees, enabling decision-makers to articulate their uncertainties, reservations, or ambiguities in their evaluations. This additional parameter better captures the inherent uncertainty present in real-world problems. Atanassov was the first who introduce the "*Intuitionistic Fuzzy Set*" (IFS). Different decision-making problems with uncertainties can be solved by using IFS.

Throughout our literature research, we found that the usage of IFS with MCDM techniques for the selection of SRs received little attention. In this work, we propose an IFS and TOPSIS-based method to perform the SRs selection. The TOPSIS is one of the most popular MCDM techniques used for the SRs selection because of its apparent simplicity and ease of implementation when compared to other MCDM methods like ELECTRE, PROMETHEE, AHP, etc., which are significantly more complicated and less transparent to the decision makers (DMs) (Sadiq et al. 2022; Nazim et al. 2023). So, using the TOPSIS technique along with IFS increases the possibility that the SRs' selection process will be successful. The IFS has proven to be very effective in dealing with uncertainty and ambiguity. It is a suitable tool for use when presenting ambiguous or imprecise decision information (Memari et al. 2019). It motivated us to expand the traditional TOPSIS method with the help of IFS to address the subjective nature, ambiguities, and complexities of software requirements selection.

The contributions to this paper are as follows:

- A proposed framework for decision-making is efficient and adaptable.
- Improved the overall quality of decision-making by leveraging the benefits of IFS alongside the TOPSIS technique.
- Illustration of the method's effectiveness using the requirements of an Institute Examination System as a case study.
- Demonstration of reduced execution times compared to the fuzzy TOPSIS method while maintaining consistency in ranking functional requirements.

The forthcoming sections of this manuscript are structured as follows: Sect. 2 presents an overview of the literature review, Sect. 3 explains the notion of Intuitionistic Fuzzy Sets (IFS) and the implementation of the intuitionistic fuzzy TOPSIS method, Sect. 4 provides a dissection of a numerical example, a comparative analysis of the proposed method with the Fuzzy TOPSIS method is presented in Sect. 5, and finally, Sect. 6 furnishes the concluding remarks alongside possibilities for future enhancements within our investigation.

2 Related work

Numerous methods for ranking Software Requirements (SRs) have been recorded in existing literature (Pitangueira et al. 2015; Achimugu et al. 2014a; Karlsson et al. 1998; Bukhsh et al. 2020), as illustrated in Table 1. For instance, Ejnioui et al. (Ejnioui et al. 2012a) introduced a method that utilized fuzzy ranking numbers for requirement prioritization. This approach incorporated anticipated values to represent the importance of various aspects within a requirement. Tonella et al. formulated an approach employing an 'interactive genetic algorithm (IGA)' (Tonella et al. 2013) that integrated knowledge acquisition with prioritization and interdependencies. This approach's validity was confirmed through a practical case study, and its results were compared with those of an incomplete AHP. Perini et al. (Perini et al. 2009) used "Case-Based Ranking (CBRank)" and AHP methods to evaluate requirements prioritization based on parameters like accuracy, time consumption, and ease of use. A collection of twenty requirements was used for the

Table 1 Pros and cons of different software requirements prioritization methods

Sources	Methodology used	Pros	Cons
Ejnioui et al. (2012a)	Fuzzy multi-attribute decision-making	• Can integrate various factors and characteristics to offer a comprehensive perspective on quality needs	• Not considering other types of requirements
Tonella et al. (2013)	An Interactive Genetic Algorithm (IGA)	• Highly efficient in selectively extracting pertinent information, such as project planning, definition of test plans, and evaluation of design and architectural options • Generates a reasonably close estimate of the ranking of reference requirements, necessitating a manageable level of manual input and accommodating a reasonable margin for human error	• Does not guarantee that the elicitation count remains within the assigned budget, as the quantity of elicited pairs is a modifiable parameter of the algorithm • There are no feasible stopping criteria for the IGA method
Achimugu et al. (2014b)	AHP and CBRank methods	• It boasts a user-friendly interface, making it simple to operate, along with minimized computational intricacies • It effectively handles concerns related to rank reversals	• Inadequate for extensive quantities of stakeholders and requirements
Achimug et al. (2014)	A scalable, innovative prioritization model	• Subjective assessments can be quantified to enhance the efficiency of comparisons and mitigate bias in the pairwise evaluation procedures	• Not executed on an actual real-life project
Achimugu et al. (2015)	Fuzzy Multicriteria Decision Making (FMCDM) approach	• This method exhibited superior performance concerning factors such as inaccuracy, significant divergence, and intricacy	• Not suitable for a large set of requirements
Ahmad et al. (2017)	Fuzzy MoSCoW	• It utilizes the MoSCoW technique within a fuzzy environment for prioritizing software requirements (SRP)	• Not suitable for a large set of requirements
Asif et al. (2017)	SAFFRON: a semi-automated framework for software requirements prioritization	• It tackles the challenge of rank reversal in prioritizing requirements • Mitigate rank reversal problems and reduce stakeholder interactions	• Does not employ an extensive collection of requirements
Babar et al. (2015)	PHandler: an intelligent system designed for an adaptable process of prioritizing software requirements	• It employs a fusion of a value-centered intelligent method for prioritizing requirements, neural network capabilities, and analytical hierarchy process to enhance the scalability of the requirement prioritization process • It is used to effectively prioritize extensive requirements for larger-scale projects	• Not suitable for a small set of requirements
Mougouei et al. (2019)	Prioritization And Partial Selection (PAPS)	• Diminishes the count of neglected security requirements by permitting partial fulfilment of said requirements • Capable of scalability to accommodate a significant volume of requirements	• Did not consider the elements of usability and performance
Dabbagh and Lee (2015)	a numerical method rooted in Fuzzy Logic and the Alpha cut technique	• Create a connection between FRs and non-NFRs to facilitate the prioritization process of NFRs	• No practical real-world illustration is utilized
Jawale et al. (2017)	Adaptive fuzzy hierarchical cumulative voting (AFHCV)	• It enhances the scope of events that can occur during runtime • It involves incorporating the addition of requirement re-prioritization. This re-prioritization is applied to enhance the outcomes of the suggested AFHCV	• Not suitable in case of removal of any requirements from the existing set
Hujainah et al. (2019)	StakeQP	• It can yield more precise outcomes while demanding less time • It is more adept at overcoming significant drawbacks like inadequate incorporation of detailed low-level implementation, time-intensive procedures, absence of automation, strong reliance on expert input, and a lack of attribute measurement criteria (AMC); for evaluating stakeholders when contrasted with alternative methods	• Apply only to a large dataset

experimental purpose, and based on the result, the CBRank was considered the best method.

A web-based tool was developed by Achimugu et al. 2014b to prioritize the SRs. Within this tool, the process encompassed normalizing and aggregating the relative weights assigned by stakeholders. In (Achimugu et al. 2014c), a method for prioritizing an extensive array of requirements is detailed, wherein the k-means algorithm was employed to manage this vast set of requirements. In an alternative investigation, Achimugu et al. (Achimug et al. 2014) introduced an effective approach to requirement prioritization, utilizing the confidence function. (Momeni et al. 2014) devised a "neuro-fuzzy system (NFS)" to prioritize the quality requirements of the software, demonstrating its efficacy using thirteen requirements from financial and sales systems. (Achimugu et al. 2015) adopted a fuzzy-based MCDM technique for prioritizing SRs, while Bajaj et al. (Bajaj and Arora 2013) utilized both the "fuzzy AHP" method and "alpha cut" operations to prioritize conflicting

requirements in the context of a "travel management planning" website.

In (Ahmad et al. 2017), a "*fuzzy MoSCoW method*" was introduced for prioritizing SRs, utilizing a library system as a case study. Asif et al. (2017) put forth a "*semi*-automated *framework*" to prioritize SRs. Babar et al. (2015) introduced the "*Priority Handler* (PHandler)" which employed three distinct approaches—namely, the "*value-based intelligent prioritization* (VIRP) method," "*neural network*," and "*AHP*"—to prioritize an extensive collection of SRs. (Mougouei et al. 2019) proposed a fuzzy logic-based technique to prioritize system requirements, demonstrated through an online banking system case study. Afrin and Sadiq (2017) suggested an integrated approach by combining "*fuzzy AHP*" and "*fuzzy TOPSIS*" for prioritization, employing an "*Institute Examination System* (IES)" as a case study. Ejnioui et al. (Ejnioui et al. 2012b) presented a "*fuzzy multi-attribute decision-making*" technique for prioritizing SRs.

Dabbagh and Lee (2015) performed the prioritization of non-functional requirements using fuzzy logic. Lima et al. (2011) introduced an approach that employs fuzzy logic to prioritize System Requirements (SRs), showcasing its effectiveness with nine requirements. Jawale et al. (2017) applied "adaptive fuzzy hierarchical cumulative voting" for SR prioritization. Achimugu et al. (2014d) formulated a model based on fuzzy TOPSIS to prevent the delivery of subpar system development due to unclear requirements. Hujainah et al. (2019) suggested a method termed "semi-automated stakeholder quantification and prioritization" for SR selection. In (Logue and McDaid 2008), the handling of uncertainty during SR prioritization was discussed. Achimugu et al. (2014e) utilized a value-based intelligent technique for SR prioritization.

The above techniques have been proposed by using “fuzzy set theory (FST)” to perform the SR prioritization, but we didn’t find any study that deals with the use of IFS for the SRs prioritization. The IFS is much better than the ordinary fuzzy set at dealing with ambiguity and uncertainty. Therefore, we developed a method using IFS and TOPSIS to find out the ranking of SRs.

Table 2 compares 25 research papers from the "International Journal of System Assurance Engineering and Management" and "Wireless Personal Communications" journals. It analyzes the papers based on their publication year, journal, application area, methodology used, and their focus within that area. Table 2 reveals diverse research areas covered, including cloud selection, network optimization (SDN), wireless sensor networks, and 5G communication. It also highlights various methodologies used, such as fuzzy logic, AHP, and simulations.

3 Materials and methods

3.1 Intuitionistic fuzzy sets

The IFS is an extension of the classical fuzzy set that aims to capture the hesitation and uncertainty in membership degrees (Atanassov 1986). Coined by Krassimir Atanassov in 1983, IFS gives a more flexible framework for expressing and reasoning with ambiguous or uncertain information. In classical fuzzy sets, A membership degree, which ranges from 0 to 1, is given to each element, which indicates how much it belongs to the set. Contrarily, IFS introduces an extra parameter called the non-membership degree, which measures how much an element does not belong to a set (Patel et al. 2023). Furthermore, IFS contains a degree of hesitation that represents the level of hesitation or uncertainty in assigning precise membership or non-membership values. Incorporating these three degrees enables intuitionistic fuzzy sets to express uncertainty and ambiguity more nuancedly, making them an effective tool for pattern recognition, decision-making, and other areas involving ambiguous or imperfect information (Boran et al. 2009; Anwar et al. 2023a, 2024, 2023b).

If X is a finite set, and F is an IFS in X, then it can be represented as Eq. (1):

$$F = \{< x, \mu_F(x), v_F(x) > | x \in X\} \tag{1}$$

where $\mu_F(x) : X \to [0,1]$ and $v_F(x) : X \to [0,1]$ represent the membership and non-membership functions, respectively, such that the sum of the membership function and the non-membership function should be in the range of 0 and 1, as illustrated by Eq. (2).

$$0 \leq \mu_F(x) + v_F(x) \leq 1 \tag{2}$$

Contrary to traditional FST, the IFS has an extra parameter, i.e., $\pi_F(x)$. This additional parameter is called as an *“intuitionistic fuzzy index”* or *“hesitation degree”* of x to represent whether it belongs to F or not (Atanassov 1986).

Let’s assume $\pi_F(x)$ is the *“hesitation degree”* over whether or not x belongs to F, then F can be represented as Eq. (3).

$$\pi_F(x) = 1 - \mu_F(x) - v_F(x) \tag{3}$$

It is obvious for every $x \in X$, such that the hesitation degree should be in the range of 0 and 1, as illustrated by Eq. (4).

$$0 \leq \pi_F(x) \leq 1 \tag{4}$$

If the value of $\pi_F(x)$ is smaller, then the knowledge of x is more certain, whereas it is uncertain if the value of $\pi_F(x)$ is

Table 2 Comparative analysis of research focus and methodologies used in different current studies

Reference	Year	Application area	Methodology used	Focus area
Sarkar and Srivastava (2024)	2024	Neutrosophic fuzzy optimization	Neutrosophic fuzzy approach	Recent development
Boutkhoum et al. (2017)	2017	Cloud selection	Fuzzy AHP-TOPSIS	Decision-making
Faiz and Daniel (2024)	2024	Cloud selection	Fuzzy logic	QoS-based multi-criteria model
Sengazhani Murugesan et al. (2020)	2020	Facility layout	AHP	Mail operational performance improvement
Sadiq and Jain (2015)	2015	Goal-oriented requirements elicitation	Fuzzy approach	Goal selection
Rakkiannan et al. (2023)	2023	Network slicing	Federated learning	SDN and NFV
Ramprasath and Seethalakshmi (2021)	2021	Network monitoring	SDN	DDoS detection and mitigation
Ruhan Bevi et al. (2019)	2019	Internet-of-Things	SDN	Gateway design
Bhardwaj and Panda (2022)	2022	SDN	RYU controller	Performance evaluation
Shamim et al. (2018)	2018	SDN	Data communication speed and fault tolerance	
Let et al. (2023)	2023	SDN routing	QoS and models	Issues and analysis
Keshari et al. (2021)	2021	SDN	Systematic review	QoS
Swami et al. (2021)	2021	SDN	TCP-SYN DDoS attack	Detection and analysis
Pakdel et al. (2022)	2022	Wireless sensor networks	Dual sinks	Environmental monitoring
Wan et al. (2020)	2020	Wireless sensor networks	Locally weighted linear regression	Comment on DV-Hop algorithm
Njoya et al. (2020)	2020	Wireless sensor networks	Hybrid deployment scheme	Connectivity and coverage
Keramatpour et al. (2017)	2017	Wireless sensor networks	Deployment of intrusion detection systems	Coverage maximization
Ding et al. (2019)	2019	Wireless sensor networks	LEACH protocol	Cluster head selection
Bouckaert et al. (2010)	2010	Wireless networks	Interconnection strategies and challenges	Mesh and sensor networks
Gupta and Singh (2023)	2023	Wireless sensor networks	Localization of sensor nodes	Performance evaluation
Liu and Li (2023)	2023	Mobile wireless sensor networks	Coverage algorithm	Perceived environment
Indoonundon and Fowdur (2022)	2022	5G New Radio	Hierarchical and statistical QAM	Error performance enhancement
Gupta et al. (2022)	2022	5G wireless communication	Simulink models for cognitive radio	Comparative study
Challa and Bagadi (2021)	2021	Beyond 5G wireless networks	MU-MIMO system design	Joint precoding and detection schemes
Balachander and Krishnan (2022)	2022	5G wireless communication	Cognitive radio network	CFO synchronization and PAPR minimization

great. The conventional fuzzy set concept is recovered when $\mu_F(x) = 1 - v_F(x)$ for all the elements of the universe set (Patel et al. 2023).

If F and G are two IFSs of the universe set X, then the multiplication operator can be illustrated as Eq. (5) (Atanassov 1986):

$$F \otimes G = \{\mu_F(x).\mu_G(x), v_F(x) + v_G(x) - v_F(x).v_G(x) | x \in X\} \quad (5)$$

The component-wise matrix multiplication can be performed as given by Eq. (6):

$$F^{\circ}G = [< \min\{\mu_F(x), \mu_G(x)\}, \max\{v_F(x), v_G(x)\} >] \quad (6)$$

3.2 Intuitionistic fuzzy TOPSIS: A proposed methodology

To understand the process of the "Intuitionistic Fuzzy TOPSIS (IFT)" method, let's consider a set of functional requirements (FRs) as $F = \{F_1, F_2, \ldots, F_p\}$ and a set of non-functional requirements (NFRs) as $N = \{N_1, N_2, \ldots, N_q\}$. The complete working procedure of the IFT method is given in Fig. 1. The proposed IFT approach has a computational complexity of O(m × n), and empirical analysis confirms a linear runtime growth with increasing requirement size, supporting its scalability for large-scale applications.

Step 1: Identifying the weights of DMs

Suppose there are d DMs in the decision group. The linguistic terms are used to express the significance of DMs. These linguistic terms are expressed in the form of IFNs.

If an IFN is to represent the rating of rth DMs is $E_r = [\mu_r, v_r, \pi_r]$, then the weight of rth DM can be calculated by using Eq. (7):

$$w_r = \frac{\left(\mu_r + \pi_r\left(\frac{\mu_r}{\mu_r+v_r}\right)\right)}{\sum_{r=1}^{d}\left(\mu_r + \pi_r\left(\frac{\mu_r}{\mu_r+v_r}\right)\right)} \tag{7}$$

and $\sum_{r=1}^{d} w_r = 1$

Step 2: Create an aggregated Intuitionistic Fuzzy Decision Matrix

Suppose $S^{(r)} = (s_{ij}{}^{(r)})_{p\times q}$ is an *"Intuitionistic Fuzzy Decision Matrix (IFDM)"* of each DM. The weight of each DM is $w = \{w_1, w_2, w_3, \ldots, w_d\}$ and $\sum_{r=1}^{d} w_r = 1, w_r = \in [0,1]$. In the process of group decision making, all the views of all DMs need to be aggregated together into an aggregated IFDM with the help of an *"Intuitionistic Fuzzy Weighted Averaging (IFWA)"* operator, given by Xu (2007). $S = (s_{ij})_{p\times q}$, where

$$\begin{aligned} s_{ij} &= IFWA_w\left(s_{ij}^{(1)}, s_{ij}^{(2)}, \ldots, s_{ij}^{(l)}\right) \\ &= w_1 s_{ij}^{(1)} \oplus w_2 s_{ij}^{(2)} \oplus w_3 s_{ij}^{(3)} \oplus \cdots \oplus w_l s_{ij}^{(d)} \\ &= \left[1 - \prod_{r=1}^{d}\left(1-\mu_{ij}^{(r)}\right)^{w_r}, \prod_{r=1}^{d}\left(v_{ij}^{(r)}\right)^{w_r}, \prod_{r=1}^{d}\left(1-\mu_{ij}^{(r)}\right)^{w_r} - \prod_{r=1}^{d}\left(v_{ij}^{(r)}\right)^{w_r}\right] \end{aligned} \tag{8}$$

where $s_{ij} = (\mu_{F_i}(x_j), v_{F_i}(x_j), \pi_{F_i}(x_j))$ $(i = 1,2,\ldots,p; j = 1,2,\ldots,q)$

By using Eq. (8), the aggregated IFDM can be represented as given below:

$$S = \begin{bmatrix} (\mu_{F_1}(x_1), v_{F_1}(x_1), \pi_{F_1}(x_1)) & (\mu_{F_1}(x_2), v_{F_1}(x_2), \pi_{F_1}(x_2)) & \ldots & (\mu_{F_1}(x_q), v_{F_1}(x_q), \pi_{F_1}(x_q)) \\ (\mu_{F_2}(x_1), v_{F_2}(x_1), \pi_{F_2}(x_1)) & (\mu_{F_2}(x_2), v_{F_2}(x_2), \pi_{F_2}(x_2)) & \ldots & (\mu_{F_2}(x_q), v_{F_2}(x_q), \pi_{F_2}(x_q)) \\ \vdots & \vdots & \ddots & \vdots \\ (\mu_{F_p}(x_1), v_{F_p}(x_1), \pi_{F_p}(x_1)) & (\mu_{F_p}(x_2), v_{F_p}(x_2), \pi_{F_p}(x_2)) & \ldots & (\mu_{F_m}(x_q), v_{F_p}(x_q), \pi_{F_p}(x_q)) \end{bmatrix}$$

$$S = \begin{bmatrix} s_{11} & s_{12} & \ldots & s_{1p} \\ s_{21} & s_{22} & \ldots & s_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ s_{q1} & s_{q2} & \ldots & s_{qp} \end{bmatrix}$$

Step 3: Computing the weights of NFRs

The views of all those who make decisions for each NFR are aggregated to compute the weight (T) for all NFRs. Suppose

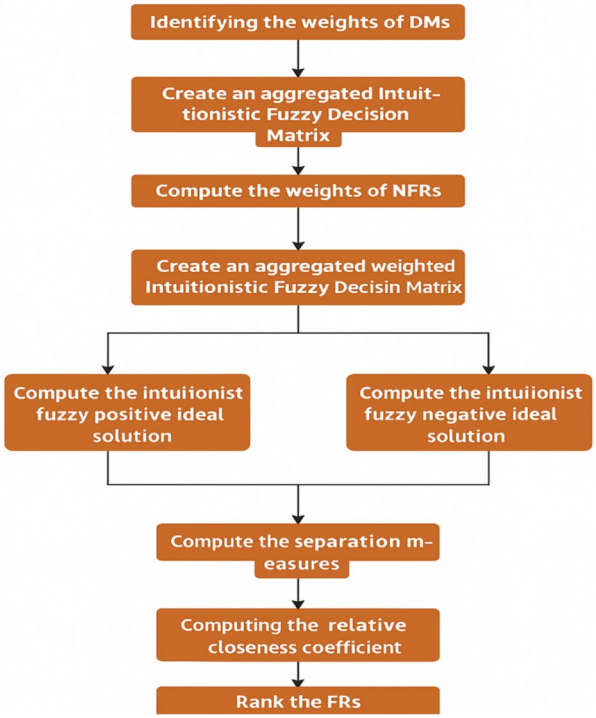


Fig. 1 Flow chart of proposed methodology

$t_j^r = \left[\mu_j^{(r)}, v_j^{(r)}, \pi_j^{(r)}\right]$ is an IFN for NFR N_j given by the rth DM, Subsequently, the IFWA operator is applied to calculate the weights of NFRs, as depicted in Eq. (9).

$$\begin{aligned} t_j &= IFWA_w\left(t_j^{(1)}, t_j^{(2)}, \ldots, t_j^{(d)}\right) \\ &= w_1 t_j^{(1)} \oplus w_2 t_j^{(2)} \oplus w_3 t_j^{(3)} \oplus \cdots \oplus w_d t_j^{(d)} 1 \end{aligned} \tag{9}$$

where $T = [t_1, t_2, t_3, \ldots, t_j]$, $t_j = (\mu_j, v_j, \pi_j)(j = 1,2,\ldots,q)$

Step 4: Generating a compounded weighted Intuitionistic Fuzzy Decision Matrix (IFDM)

Once the aggregated IFDM is formed and the weight of the NFRs (T) is calculated, the aggregated weighted IFDM is created by using Eq. (10) and Eq. (11).

$$S \otimes T = \{< x, \mu_{F_i}(x) \bullet \mu_S(x), v_{F_i}(x) + v_T(x) - v_{F_i}(x) \bullet v_T(x) > | x \in X\} \tag{10}$$

and

Table 3 Linguistic terms utilized to evaluate the significance of DMs and NFRs

Linguistic variables	IFNs
Very high (VH)	(0.90,0.10)
High (H)	(0.75,0.20)
Medium (M)	(0.50,0.45)
Low (L)	(0.35,0.60)
Very low (VL)	(0.10,0.90)

$$\pi_{F_i \bullet T(x)} = 1 - v_{F_i}(x) - v_T(x) - \mu_{F_i}(x) \bullet \mu_T(x) + v_{F_i}(x) \bullet v_T(x) \tag{11}$$

The aggregated weighted IFDM can be described as follows:

$$S\prime = \begin{bmatrix} (\mu_{F_1T}(x_1), v_{F_1T}(x_1), \pi_{F_1T}(x_1)) & (\mu_{F_1T}(x_2), v_{F_1T}(x_2), \pi_{F_1T}(x_2)) & \cdots & (\mu_{F_1T}(x_q), v_{F_1T}(x_q), \pi_{F_1T}(x_q)) \\ (\mu_{F_2T}(x_1), v_{F_2T}(x_1), \pi_{F_2T}(x_1)) & (\mu_{F_2T}(x_2), v_{F_2T}(x_2), \pi_{F_2T}(x_2)) & \cdots & (\mu_{F_2T}(x_q), v_{F_2T}(x_q), \pi_{F_2T}(x_q)) \\ \vdots & \vdots & \ddots & \vdots \\ (\mu_{F_pT}(x_1), v_{F_pT}(x_1), \pi_{F_pT}(x_1)) & (\mu_{F_pT}(x_2), v_{F_pT}(x_2), \pi_{F_pT}(x_2)) & \cdots & (\mu_{F_pT}(x_q), v_{F_pT}(x_q), \pi_{F_pT}(x_q)) \end{bmatrix}$$

$$S' = \begin{bmatrix} s'_{11} & s'_{12} & \cdots & s'_{1p} \\ s'_{21} & s'_{22} & \cdots & s'_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ s'_{q1} & s'_{q2} & \cdots & s'_{qp} \end{bmatrix}$$

$s'_{ij} = \left(\mu'_{ij}, v'_{ij}, \pi'_{ij}\right) = \left(\mu_{F_iT}(x_j), v_{F_iT}(x_j), \pi_{F_iT}(x_j)\right)$ is an element of the aggregated weighted IFDM.

Step 5: Calculating the intuitionistic fuzzy positive ideal solution (FPIS) and the fuzzy negative ideal solution (FNIS)

If G_1 and G_2 are benefit NFR and cost NFR, respectively, FPIS (i.e., F^*) and FNIS (i.e., F^-) are computed by using Eq. (12) and Eq. (13).

$$F^* = (\mu_{F^*T}(x_j), v_{F^*T}(x_j)) \tag{12}$$

$$F^- = (\mu_{F^-T}(x_j), v_{F^-T}(x_j)) \tag{13}$$

where, the values of $\mu_{F^*T}(x_j)$, $v_{F^*T}(x_j)$, $\mu_{F^-T}(x_j)$, and $v_{F^-T}(x_j)$ are computed by using Eq. (14), Eq. (15), Eq. (16), and Eq. (17), respectively.

$$\mu_{F^*T}(x_j) = \left(\left(\max_i \mu_{F_i \bullet T}(x_j) \,|\, j \in G_1\right), \left(\min_i \mu_{F_i \bullet T}(x_j) \,|\, j \in G_2\right)\right) \tag{14}$$

$$v_{F^*T}(x_j) = \left(\left(\min_i v_{F_i \bullet T}(x_j) \,|\, j \in G_1\right), \left(\max_i v_{F_i \bullet T}(x_j) \,|\, j \in G_2\right)\right) \tag{15}$$

$$\mu_{F^-T}(x_j) = \left(\left(\min_i \mu_{F_i \bullet T}(x_j) \,|\, j \in G_1\right), \left(\max_i \mu_{F_i \bullet T}(x_j) \,|\, j \in G_2\right)\right) \tag{16}$$

$$v_{F^-T}(x_j) = \left(\left(\max_i v_{F_i \bullet T}(x_j) \,|\, j \in G_1\right), \left(\min_i v_{F_i \bullet T}(x_j) \,|\, j \in G_2\right)\right) \tag{17}$$

Step 6: Computing the separation measures

Distance metrics such as "Hamming distance" and "Euclidean distance" measurements can be employed to quantify the dissimilarity between functional requirements (FRs) within the context of Intuitionistic Fuzzy Sets (IFS) (Atanassov 2016; Szmidt and Kacprzyk 2000; Grzegorzewski 2004). The calculation of the separaton measure of each FR from intuitionistic FPIS (i.e., SM_{i^*}), and separation measure of each FR from intuitionistic FNIS (i.e., SM_{i^-}) is performed after the distance measure has been choosen. In our work, we used normalized Euclidean distance (Szmidt and Kacprzyk 2000) as illustrated in Eq. (18).

$$SM^* = \sqrt{\frac{1}{2q}\sum_{j=1}^{q}\left[\left(\mu_{F_i \bullet T}(x_j) - \mu_{F^*T}(x_j)\right)^2 + \left(v_{F_i \bullet T}(x_j) - v_{F^*T}(x_j)\right)^2 + \left(\pi_{F_i \bullet T}(x_j) - \pi_{F^*T}(x_j)\right)^2\right]} \tag{18}$$

$$SM^- = \sqrt{\frac{1}{2q}\sum_{j=1}^{q}\left[\left(\mu_{F_i \bullet T}(x_j) - \mu_{F^-T}(x_j)\right)^2 + \left(v_{F_i \bullet T}(x_j) - v_{F^-T}(x_j)\right)^2 + \left(\pi_{F_i \bullet T}(x_j) - \pi_{F^-T}(x_j)\right)^2\right]} \tag{19}$$

Table 4 Linguistic terms used to assess the significance of the FRs

Linguistic variables	IFNs
Extremely important (EI)	(1.00,0.00)
Very very important (VVI)	(0.90,0.10)
Very important (VI)	(0.80,0.10)
Important (I)	(0.70,0.20)
Medium important (MI)	(0.60,0.30)
Medium (M)	(0.50,0.40)
Medium unimportant (MI)	(0.40,0.50)
Unimportant (U)	(0.25,0.60)
Very unimportant (VU)	(0.10,0.75)
Very very unimportant (VVU)	(0.10,0.90)

Step 7: Calculating the relative closeness coefficient (CC) with respect to the intuitionistic ideal solution

The relative CC of an FR, denoted as F_i, In relation to the ideal FR F*, it can be computed using Eq. (20).

$$CC_{i^*} = \frac{SM_{i^-}}{SM_{i^*} + SM_{i^-}} \tag{20}$$

Table 5 Significance of DMs and their corresponding weights

	DM_1	DM_2	DM_3
Linguistic variables	M	VH	H
Weights	0.5263	0.9000	0.7895

where $0 \leq CC_{i^*} \leq 1$

Step 8: Rank the FRs

After computing the relative CC of all the FRs, the ranking order of FRs is found by arranging the values of closeness coefficients of all the FRs in the ascending/descending order, i.e., the FR with the highest value of CC_{i^*} is considered the highest-ranked FR, whereas the FR with the lowest value of CC_{i^*}. It is considered the lowest-ranked FR.

4 A critical case study: results and discussion

Within the software development process, accommodating all elicited requirements for a single software product release proves unfeasible. Hence, the task of choosing the most critical SRs from an extensive set of gathered SRs becomes imperative. This study focused on ten distinct FRs and four NFRs pertaining to an Institute Examination System (IES). A group comprising three DMs was convened to evaluate Functional Requirements (FRs) in accordance with

Table 6 Weight of the NFRs

NFRs	DM_1	DM_2	DM_3
nfr_1	VH	H	VH
nfr_2	H	VH	L
nfr_3	H	M	M
nfr_4	M	VH	M

Table 7 Rating of NFRs based on IFNs

NFRs	DM_1	DM_2	DM_3
nfr_1	(0.90,0.10)	(0.75,0.20)	(0.90,0.10)
nfr_2	(0.75,0.20)	(0.90,0.10)	(0.35,0.60)
nfr_3	(0.75,0.20)	(0.50,0.45)	(0.50,0.45)
nfr_4	(0.50,0.45)	(0.90,0.10)	(0.50,0.45)

NFRs. The assessment of FRs has been grounded on the subsequent four NFRs:

nfr_1: Security.
nfr_2: Reliability.
nfr_3: Usability.
nfr_4: Budget.

The steps of the process of selection of SRs are as follows:

Step 1: Identifying the weights of DMs.

Table 3 displays the linguistic variables employed for assessing the importance of DMs and NFRs, while Table 4 illustrates the linguistic variables used to assess the significance of FRs. Table 5 outlines the extent of DMs' influence on group decisions. The weights assigned to all DMs are calculated using Eq. (7).

$$w_{DM_1} = \frac{\left(0.50 + (1 - 0.50 - 0.45)\left(\frac{0.50}{0.50+0.45}\right)\right)}{0.9 + \left(0.50 + 0.05 * \frac{0.50}{0.95}\right) + \left(0.75 + 0.05 * \frac{0.75}{0.95}\right)} = \frac{0.50 + 0.05 * 0.526}{0.9 + 0.526 + 0.789} = 0.2376$$

$$w_{DM_2} = \frac{\left(0.90 + (1 - 0.90 - 0.10)\left(\frac{0.90}{0.90+0.10}\right)\right)}{0.9 + \left(0.50 + 0.05 * \frac{0.50}{0.95}\right) + \left(0.75 + 0.05 * \frac{0.75}{0.95}\right)} = \frac{0.90}{0.9 + 0.526 + 0.789} = 0.4063$$

$$w_{DM_3} = \frac{\left(0.75 + (1 - 0.75 - 0.20)\left(\frac{0.75}{0.75+0.20}\right)\right)}{0.9 + \left(0.50 + 0.05 * \frac{0.50}{0.95}\right) + \left(0.75 + 0.05 * \frac{0.75}{0.95}\right)} = \frac{0.75 + 0.05 * 0.789}{0.9 + 0.526 + 0.789} = 0.3564$$

Table 8 Aggregated weights of NFRs

NFRs	nfr_1	nfr_2	nfr_3	nfr_4
Aggregated weights	(0.839,0.141,0.020)	(0.668,0.302,0.030)	(0.559,0.391,0.050)	(0.662,0.308,0.030)

Table 9 Ratings of the FRs

NFRs	FRs	DMs		
		DM_1	DM_2	DM_3
nfr_1	fr_1	EI	MI	I
	fr_2	VI	I	EI
	fr_3	MI	I	EI
	fr_4	EI	EI	EI
	fr_5	I	U	VI
	fr_6	I	VVI	VI
	fr_7	EI	VVI	M
	fr_8	EI	EI	VI
	fr_9	EI	VI	MI
	fr_{10}	M	VVI	VVI
nfr_2	fr_1	VVI	EI	EI
	fr_2	M	I	M
	fr_3	M	VI	I
	fr_4	EI	VI	I
	fr_5	VVI	MI	EI
	fr_6	EI	EI	I
	fr_7	VI	VVI	I
	fr_8	U	EI	MI
	fr_9	MI	M	EI
	fr_{10}	EI	EI	EI
nfr_3	fr_1	MI	EI	I
	fr_2	EI	VVI	U
	fr_3	VVI	EI	EI
	fr_4	I	VVI	MI
	fr_5	I	VVI	EI
	fr_6	MI	EI	EI
	fr_7	U	VI	VVI
	fr_8	VVI	MI	EI
	fr_9	I	I	U
	fr_{10}	EI	M	MI
nfr_4	fr_1	EI	M	VI
	fr_2	EI	EI	VI
	fr_3	VVI	EI	EI
	fr_4	EI	I	VVI
	fr_5	I	I	M
	fr_6	U	I	M
	fr_7	M	U	VVI
	fr_8	VI	I	EI
	fr_9	I	VI	VVI
	fr_{10}	U	EI	I

Table 10 Ratings of the FRs based on IFNs

NFRs	FRs	DMs		
		DM_1	DM_2	DM_3
nfr_1	fr_1	(1.00,0.00)	(0.60,0.30)	(0.70,0.20)
	fr_2	(0.80,0.10)	(0.70,0.20)	(1.00,0.00)
	fr_3	(0.60,0.30)	(0.70,0.20)	(1.00,0.00)
	fr_4	(1.00,0.00)	(1.00,0.00)	(1.00,0.00)
	fr_5	(0.70,0.20)	(0.25,0.60)	(0.80,0.10)
	fr_6	(0.70,0.20)	(0.90,0.10)	(0.80,0.10)
	fr_7	(1.00,0.00)	(0.90,0.10)	(0.50,0.40)
	fr_8	(1.00,0.00)	(1.00,0.00)	(0.80,0.10)
	fr_9	(1.00,0.00)	(0.80,0.10)	(0.60,0.30)
	fr_{10}	(0.50,0.40)	(0.90,0.10)	(0.90,0.10)
nfr_2	fr_1	(0.90,0.10)	(1.00,0.00)	(1.00,0.00)
	fr_2	(0.50,0.40)	(0.70,0.20)	(0.50,0.40)
	fr_3	(0.50,0.40)	(0.80,0.10)	(0.70,0.20)
	fr_4	(1.00,0.00)	(0.80,0.10)	(0.70,0.20)
	fr_5	(0.90,0.10)	(0.60,0.30)	(1.00,0.00)
	fr_6	(1.00,0.00)	(1.00,0.00)	(0.70,0.20)
	fr_7	(0.80,0.10)	(0.90,0.10)	(0.70,0.20)
	fr_8	(0.25,0.60)	(1.00,0.00)	(0.60,0.30)
	fr_9	(0.60,0.30)	(0.50,0.40)	(1.00,0.00)
	fr_{10}	(1.00,0.00)	(1.00,0.00)	(1.00,0.00)
nfr_3	fr_1	(0.60,0.30)	(1.00,0.00)	(0.70,0.20)
	fr_2	(1.00,0.00)	(0.90,0.10)	(0.25,0.60)
	fr_3	(0.90,0.10)	(1.00,0.00)	(1.00,0.00)
	fr_4	(0.70,0.20)	(0.90,0.10)	(060,0.30)
	fr_5	(0.70,0.20)	(0.90,0.10)	(1.00,0.00)
	fr_6	(0.60,0.30)	(1.00,0.00)	(1.00,0.00)
	fr_7	(0.25,0.60)	(0.80,0.10)	(0.90,0.10)
	fr_8	(0.90,0.10)	(0.60,0.30)	(1.00,0.00)
	fr_9	(0.70,0.20)	(0.70,0.20)	(0.25,0.60)
	fr_{10}	(1.00,0.00)	(0.50,0.40)	(0.60,0.30)
nfr_4	fr_1	(1.00,0.00)	(0.50,0.40)	(0.80,0.10)
	fr_2	(1.00,0.00)	(1.00,0.00)	(0.80,0.10)
	fr_3	(0.90,0.10)	(1.00,0.00)	(1.00,0.00)
	fr_4	(1.00,0.00)	(0.70,0.20)	(0.90,0.10)
	fr_5	(0.70,0.20)	(0.70,0.20)	(0.50,0.40)
	fr_6	(0.25,0.60)	(0.70,0.20)	(0.50,0.40)
	fr_7	(0.50,0.40)	(0.25,0.60)	(0.90,0.10)
	fr_8	(0.80,0.10)	(0.70,0.20)	(1.00,0.00)
	fr_9	(0.70,0.20)	(0.80,0.10)	(0.90,0.10)
	fr_{10}	(0.25,0.60)	(1.00,0.00)	(0.70,0.20)

Table 11 Aggregated IFDM

FRs	NFRs			
	NFR_1	NFR_2	NFR_3	NFR_4
fr_1	(0.731,0.193,0.076)	(0.976,0.024,0.000)	(0.798,0.143,0.059)	(0.726,0.198,0.076
fr_2	(0.831,0.105,0.064)	(0.581,0.319,0.100)	(0.692,0.254,0.053)	(0.929,0.036,0.036)
fr_3	(0.783,0.152,0.064)	(0.693,0.207,0.100)	(0.976,0.024,0.000)	(0.976,0.024,0.000)
fr_4	(1.000,0.000,0.000)	(0.812,0.112,0.076)	(0.746,0.195,0.059)	(0.843,0.117,0.041)
fr_5	(0.553,0.327,0.120)	(0.814,0.146,0.041)	(0.888,0.088,0.024)	(0.629,0.271,0.100)
fr_6	(0.817,0.124,0.059)	(0.893,0.071,0.036)	(0.905,0.071,0.024)	(0.522,0.366,0.112)
fr_7	(0.781,0.183,0.036)	(0.805,0.136,0.059)	(0.705,0.219,0.076)	(0.541,0.374,0.085)
fr_8	(0.929,0.036,0.036)	(0.679,0.249,0.071)	(0.814,0.146,0.041)	(0.831,0.105,0.064)
fr_9	(0.776,0.148,0.076)	(0.702,0.234,0.064)	(0.540,0.343,0.118)	(0.812,0.124,0.064)
fr_{10}	(0.805,0.171,0.024)	(1.000,0.000,0.000)	(0.654,0.269,0.076)	(0.715,0.214,0.071)

Table 12 Aggregated weighted IFDM

FRs	NFRs			
	NFR_1	NFR_2	NFR_3	NFR_4
fr_1	(0.613,0.307,0.080)	(0.653,0.318,0.029)	(0.446,0.477,0.076)	(0.481,0.445,0.074)
fr_2	(0.697,0.231,0.072)	(0.389,0.524,0.087)	(0.387,0.546,0.067)	(0.615,0.332,0.052)
fr_3	(0.657,0.272,0.071)	(0.463,0.446,0.090)	(0.546,0.405,0.049)	(0.647,0.324,0.029)
fr_4	(0.839,0.141,0.020)	(0.543,0.380,0.077)	(0.417,0.509,0.073)	(0.558,0.389,0.053)
fr_5	(0.464,0.421,0.115)	(0.544,0.404,0.053)	(0.497,0.444,0.059)	(0.417,0.496,0.088)
fr_6	(0.685,0.247,0.068)	(0.597,0.352,0.051)	(0.506,0.434,0.060)	(0.346,0.561,0.093)
fr_7	(0.656,0.298,0.046)	(0.538,0.397,0.065)	(0.394,0.524,0.082)	(0.358,0.567,0.075)
fr_8	(0.779,0.171,0.049)	(0.454,0.476,0.070)	(0.455,0.479,0.065)	(0.550,0.381,0.069)
fr_9	(0.651,0.267,0.081)	(0.469,0.465,0.066)	(0.302,0.599,0.099)	(0.538,0.393,0.069)
fr_{10}	(0.675,0.288,0.037)	(0.668,0.302,0.030)	(0.366,0.555,0.079)	(0.474,0.456,0.071)

Table 13 Separation measures and the relative CC of each FR

FRs	SM^*	SM^-	$CC_i{}^*$	*Rank*
fr_1	0.128	0.171	0.570	5th
fr_2	0.205	0.115	0.359	9th
fr_3	0.185	0.147	0.444	8th
fr_4	0.128	0.196	0.606	2nd
fr_5	0.183	0.154	0.456	7th
fr_6	0.077	0.217	0.738	1st
fr_7	0.125	0.178	0.588	3rd
fr_8	0.148	0.168	0.533	6th
fr_9	0.190	0.105	0.356	10th
fr_{10}	0.129	0.179	0.580	4th

Step 2: Creating an aggregated IFDM depending on the views given by DMs.

DMs used linguistic variables as given in Table 3 to give the rating to each FR based on each NFR. The aggregated weights of NFRs are given in Table 6. Table 7 shows the ratings of ten FRs as linguistic variables, whereas Table 8 presents the ratings of FRs in terms of intuitionistic fuzzy numbers.

By aggregating these ratings, the aggregated IFDM was formed, as shown in Table 9.

Step 3: Computing the weights of NFRs.

The importance of NFRs in the form of the linguistic variables and corresponding IFNs is given in Table 5 and Table 10, respectively. By using Eq. (9), the views of DMs on the NFRs were compiled to form the aggregated weights, as represented in Table 11.

Step 4: Creating an aggregated weighted IFDM.

Once the weights for NFRs and the ratings for FRs were computed, the amalgamated and weighted IFDM was constructed using Eq. (10), as demonstrated in Table 12.

Step 5: Computing the intuitionistic FPIS and FNIS.

To evaluate the FRs, the NFRs can be used as the criteria. In our work, all four selected NFRs are categorized into two categories, named G_1 and G_2. The three NFRs, i.e., *"Security"*, *"Reliability"*, and *"Usability"*, have been considered as the benefit criteria ($G_1 = \{nfr_1, nfr_2, nfr_3\}$), whereas the budget NFR has been considered as a cost criterion (i.e., $G_2 = \{nfr_4\}$). The intuitionistic FPIS and FNIS values

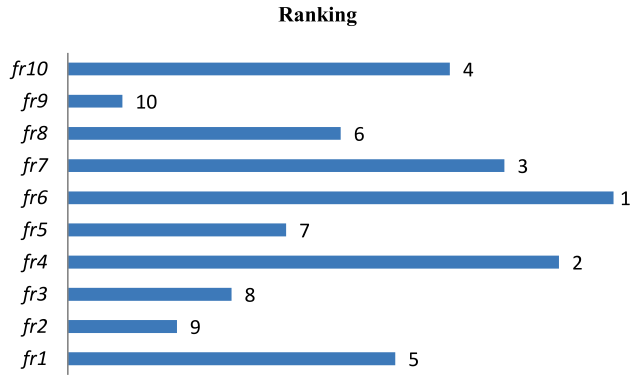


Fig. 2 Ranking order of functional requirements

were determined through the application of equations ranging from (12) to (17). The resultant calculated values for the intuitionistic fuzzy positive ideal solutions and intuitionistic fuzzy negative ideal solutions are presented as follows:

$$F^{*} = \{(0.839, 0.141, 0.020), (0.668, 0.302, 0.030), (0.546, 0.405, 0.049), (0.346, 0.567, 0.087)\}$$

$$F^{-} = \{(0.464, 0.421, 0.115), (0.389, 0.524, 0.087), (0.302, 0.599, 0.099), (0.647, 0.324, 0.029)\}$$

Step 6: Computing the separation measures.

The computation of the *"positive separation measure"* (i.e., SM^{*}) and *"negative separation measure"* (i.e., SM^{-}) is based on *"normalized Euclidean distance"* for each FR. Both the positive and negative separation measures for each FR have been computed by using Eq. (18) and Eq. (19), as shown in Table 13.

Step 7: Computing the relative CC to the intuitionistic ideal solution.

The calculation of relative correlation coefficients (CCs) for the FRs was performed using Eq. (20). Table 13 presents the specific relative CC values for ten distinct FRs.

Step 8: Rank the *FRs*.

The arrangement of FRs in terms of ranking was determined by analyzing the relative CC values associated with each FR, as outlined in Table 13. The FR having the greatest relative CC value was designated as the top-ranked FR, while the one with the lowest relative CC value was deemed the lowest-ranked, as depicted in Fig. 2.

This study utilizes the IFDM method to prioritize crucial SRs for an IES. The approach incorporates the subjective judgments of decision-makers alongside quantitative data. It involves assigning weights to decision-makers and non-functional requirements, constructing an aggregated fuzzy decision matrix, and computing separation measures to rank the SRs. This method aids in selecting the most critical SRs while considering both qualitative and quantitative factors. The proposed method's rankings were validated against expert-derived rankings (Spearman's $\rho = 0.94$) and compared with Fuzzy AHP ($\rho = 0.91$) and Fuzzy TOPSIS ($\rho = 0.96$), confirming its consistency and reliability.

5 Comparative analysis

The superiority of the IFT method over Fuzzy TOPSIS stems from its ability to handle non-membership degrees. This characteristic renders it a suitable tool for capturing the inherent uncertainty within decisions made by diverse decision makers. The experimentation utilized a system setup comprising an Intel(R) Core (TM) i3-6006U CPU operating at 2.00 GHz, 4.00 GB of RAM, and a 64-bit Windows Operating System. In terms of the ranking sequence, no significant differences were noted between the ranking sequences generated by the Fuzzy TOPSIS and the newly introduced approach. The optimal and least favorable FRs, namely FR6 and FR9, respectively, are consistent in the ranking sequence produced by both the Fuzzy TOPSIS and the introduced approach. The main difference that we have noticed was the execution time. The total execution time of Fuzzy TOPSIS was 15.62 ms, while the proposed method executed in only 13.09 ms, as shown in Table 14. The comparative analysis has been narrowed down so that it mainly deals with MCDM applications in software requirements prioritization, and other areas that are not relevant were eliminated or otherwise justifiable to give relevance to the methodology.

Intuitionistic Fuzzy TOPSIS emerges as a superior method compared to Fuzzy TOPSIS. This is due to its ability to handle both membership and non-membership degrees, providing a more nuanced representation of uncertainty. Consequently, Intuitionistic Fuzzy TOPSIS offers potentially better decision-making capabilities and may exhibit

Table 14 Comparison of the proposed method with the Fuzzy TOPSIS method

Feature	Fuzzy TOPSIS	Intuitionistic fuzzy TOPSIS	Reference
Deals with membership Degree	Yes	Yes	Sarkar and Srivastava (2024; Boutkhoum et al. (2017)
Deals with non-membership degree	No	Yes	Boutkhoum et al. (2017)
Representation of vagueness	Good	Better	(Sarkar and Srivastava 2024; Sadiq and Jain 2015)
Best functional requirement	FR6	FR6	Proposed method
Worst functional requirement	FR9	FR9	Proposed method
Total execution time (ms)	15.62	13.09	Proposed method

faster execution times. However, the choice between these methods ultimately depends on the complexity required to represent uncertainty within the specific decision-making process.

6 Conclusion and future directions

In conclusion, this study introduced a novel approach for software requirements selection using IFS and the TOPSIS technique. The motivation behind this work was to deal with the limitations of traditional methods to handle the ambiguity and uncertainty in the software requirements selection process. A thorough literature review made it clear that the existing approaches often ignore the uncertainties and hesitations associated with decision-making. The proposed method incorporates the concepts of MCDM techniques with IFS to provide a more comprehensive and reliable framework. The benefit of using IFS is its ability to capture uncertainty more effectively by considering membership, non-membership, and hesitation degrees. By considering these factors, the proposed method helps decision-makers make more informed and reliable decisions during the software requirements selection. For the assessment purpose, the ratings of FRs based on each NFR and the weights of each NFR have been used in terms of linguistic variables, and the corresponding IFNs of linguistic variables have been used for the computation purpose. After creating the aggregated weighted IFDM, the intuitionistic FPIS and intuitionistic FNIS have been computed. Finally, the calculation of relative CC s for all FRs, the work has been performed. The ranking sequence of FRs was established by evaluating their relative correlation coefficients. When we juxtaposed the IFT method with the Fuzzy TOPSIS method, it became evident that the IFT method outperformed the Fuzzy TOPSIS method on two fronts: effectively capturing the uncertainty inherent in decisions from different decision makers and minimizing the overall execution time. A potential limitation of this study lies in its neglect of the interdependencies between the NFRs. Accounting for these interrelationships could influence the final rankings. Future research could address this gap by employing established methods like the AHP or the ANP to comprehensively evaluate the interconnectedness of the NFRs, potentially leading to more nuanced prioritization of software requirements. This work also opens several new directions for software requirements selection research in the future. Here are some possible future scopes that can be addressed:

- The suggested approach can be further improved to address dynamic requirements scenarios, wherein the requirements undergo changes and developments over a period.
- Enhancing the proposed approach with machine learning techniques can augment the effectiveness and precision of selecting software requirements.

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Declarations

Conflict of interest The authors declare that they have no competing interests.

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